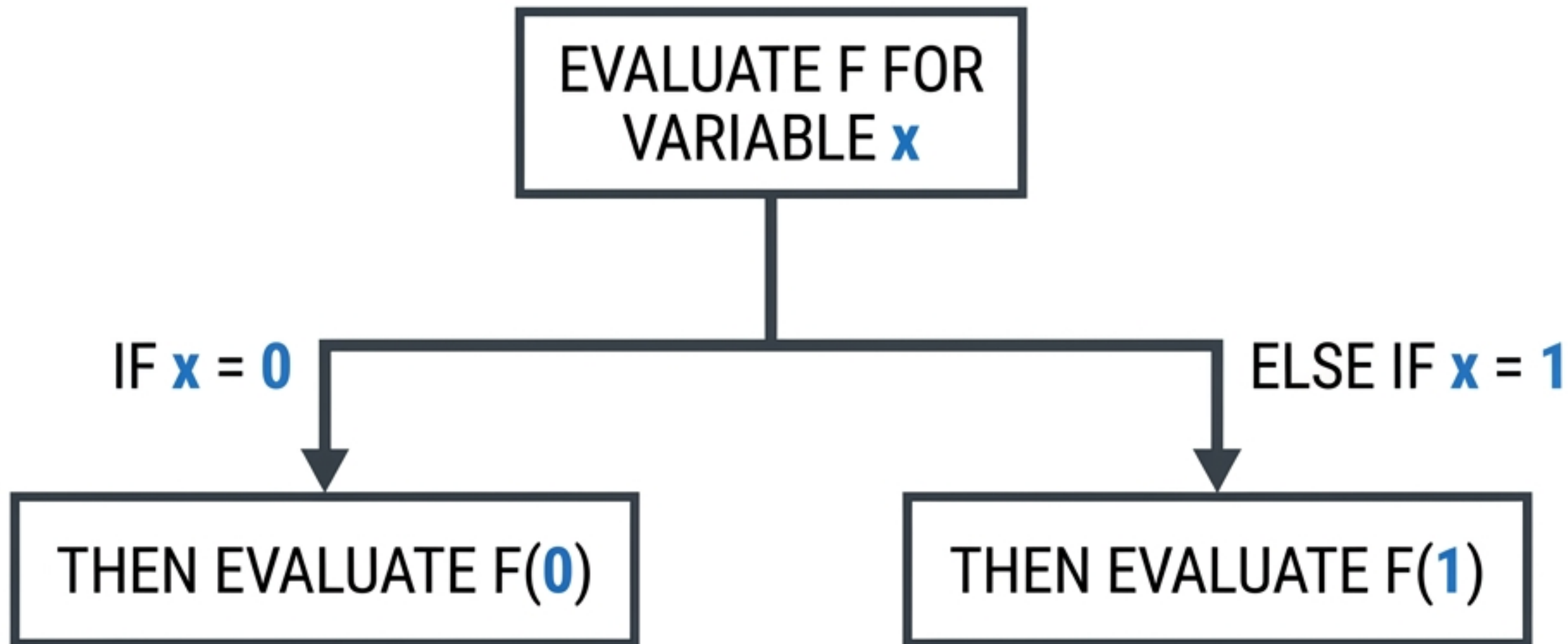


SHANNON EXPANSION THEOREM

DECOMPOSING BOOLEAN FUNCTIONS

The theorem isolates **1** variable from a function of N variables. It mathematically exposes an inherent IF THEN ELSE logic structure within any Boolean expression.



MATHEMATICAL FORMALIZATION

STANDARD FORM

SUM OF PRODUCTS

$$F = x' \cdot F(0) + x \cdot F(1)$$

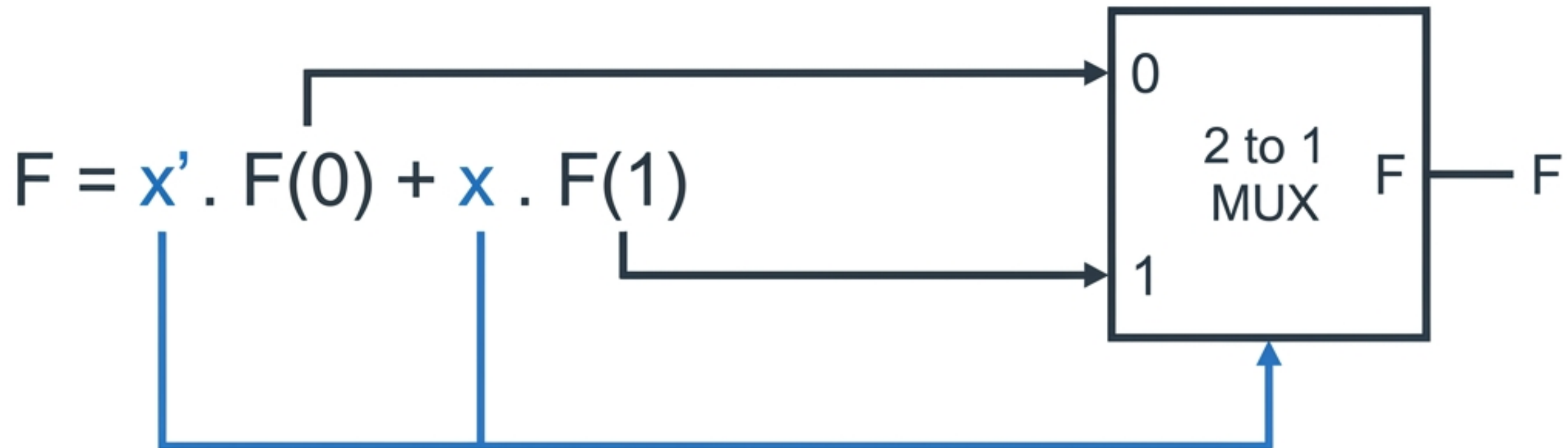
DUAL FORM

PRODUCT OF SUMS

$$F = [x' + F(1)] \cdot [x + F(0)]$$

HARDWARE IMPLEMENTATION

The expansion variable acts perfectly as the select line.
The evaluated functions $F(0)$ and $F(1)$ map exactly to the 2 data inputs.



STEP BY STEP DERIVATION

EXPAND FUNCTION USING w_1

STEP 1: STATE FUNCTION

$$= F = w_1 \cdot w_2 + w_1 \cdot w_3 + w_2 \cdot w_3$$

STEP 2: EVALUATE FOR $w_1 = 0$

$$= F(0) = 0 \cdot w_2 + 0 \cdot w_3 + w_2 \cdot w_3$$
$$F(0) = w_2 \cdot w_3$$

STEP 3: EVALUATE FOR $w_1 = 1$

$$= F(1) = 1 \cdot w_2 + 1 \cdot w_3 + w_2 \cdot w_3$$
$$F(1) = w_2 + w_3 + w_2 \cdot w_3$$
$$F(1) = w_2 + w_3$$

STEP 4: COMBINE TERMS

$$= F = w_1' \cdot F(0) + w_1 \cdot F(1)$$
$$F = w_1' \cdot (w_2 \cdot w_3) + w_1 \cdot (w_2 + w_3)$$